Study reference: forward-LLP setup and provided data

This reference collects the fixed inputs for the forward sensitivity study defined in the task. The physics model is stated in the task prompt, and the decays are to be derived from it. Everything here is the part of the setup that is *not* derived: the toy experimental configuration and the provided production data. All values are synthetic and are to be used exactly as given.

Experimental configuration

Quantity	Value	Role
Beam	pp, $\sqrt{s} = 13.6$ TeV	set in `pythia_cards/`
σ_{inel}	80.0 mb	flux weight per inelastic collision
N_{int}	2.7×10^{18}	number of primary interactions
Sensitivity rule	$N_{\text{sig}} = 3$, zero background	defines the reach
Geometry	`geometry/geometry.yaml`	decay volume, detector plane, acceptance

The normalization is the pair $(\sigma_{\text{inel}}, N_{\text{int}})$: the harvested parent flux is weighted per inelastic collision using σ_{inel}, and the final yield is scaled to N_{int} primary interactions. Two searches are defined, a two-track dimuon search and a collinear diphoton search, with acceptances specified in the geometry file. Each mass is evaluated under the search whose decay channel is open there.

Forward flux generation

The forward parent flux is generated with two Pythia cards in `pythia_cards/`, run as *separate* samples because their process types cannot be combined in a single run:

- `forward_flux.cmnd` — soft inelastic QCD, which produces the light mesons and open charm (K, D, D_s, ρ, ω, $\phi(1020)$). Its generated cross section is the inelastic cross section σ_{inel}.
- `forward_onia.cmnd` — prompt charmonium (J/ψ, $\psi(2S)$) through NRQCD, a hard process that must be generated on its own and normalized by its own generated cross section.

In both samples the listed parents are made stable and harvested at their production momenta, within a forward angular acceptance $\theta < \theta_{\max}$ about the beam axis with

$$\theta_{\max} = 0.05 \text{ rad}.$$

This is the acceptance with which the forward flux is defined; it is wider than the downstream detector solid angle, which the geometry file applies exactly, so the reported yields do not depend on the precise value above the detector opening. Each sample is weighted to a per-inelastic-collision flux using its own generated cross section, and the two are then combined.

Provided production

Production of ϕ is provided as a data product; the production matrix elements are not to be re-derived. For each open channel, per parent meson:

- `production/kappa.csv` — the g^2-stripped branching $\kappa = \mathrm{Br}(\text{parent} \to \cdots \phi)$ at $g = 1$, per (parent, m_ϕ). A parent contributes ϕ with per-collision weight $w_{\text{parent}}\,\kappa\,g^2$, where w_{parent} is its weight in the forward flux.
- `spectra/spectrum_⟨parent⟩_m⟨m_φ⟩.csv` — the ϕ energy spectrum in the parent rest frame, tabulated as (x, pdf) with $x = 2E^\star_\phi/m_{\text{parent}}$ (shape only). To place one ϕ: draw x from this distribution, take the ϕ direction isotropic in the parent rest frame, and boost to the lab with the parent momentum.

Production channels and accessible mass range

The scalar is radiated from a parent meson only below the endpoint of the $(\phi + X)$ phase space, where X collects the other final-state particles,

$$m_\phi \;<\; m_{\text{parent}} - \sum_{i \in X} m_i.$$

For the two production topologies, with the neutrino taken massless, this is

$$M \to \mu\,\nu\,\phi: \quad m_\phi < m_M - m_\mu, \qquad V \to \mu^+\mu^-\phi: \quad m_\phi < m_V - 2m_\mu.$$

The resulting per-parent endpoints are:

Parent	Topology	m_{parent} [GeV]	m_ϕ endpoint [GeV]
K	$M \to \mu\nu\phi$	0.494	0.388
D	$M \to \mu\nu\phi$	1.870	1.764
D_s	$M \to \mu\nu\phi$	1.968	1.863
ρ	$V \to \mu\mu\phi$	0.775	0.564
ω	$V \to \mu\mu\phi$	0.783	0.571
$\phi(1020)$	$V \to \mu\mu\phi$	1.019	0.808
J/ψ	$V \to \mu\mu\phi$	3.097	2.886
$\psi(2S)$	$V \to \mu\mu\phi$	3.686	3.475

The accessible range extends up to the heaviest parent's endpoint. The study is evaluated on the pinned mass grid [GeV]

$$0.06,\ 0.12,\ 0.18,\ 0.205,\ 0.23,\ 0.28,\ 0.34,\ 0.42,\ 0.70,\ 1.20,\ 1.90,\ 2.30,$$

and the rows present in `kappa.csv` are exactly the channels open at each mass.

Probe points

Report the following scalar probes:

Field	Probe point	Quantity
`width_probe_gev`	$m_\phi = 0.30,\ g = 4.2 \times 10^{-4}$	$\Gamma(\phi \to \mu\mu)$ [GeV]
`width_gg_probe_gev`	$m_\phi = 0.12,\ g = 1.0 \times 10^{-3}$	$\Gamma(\phi \to \gamma\gamma)$ [GeV]
`ctau_probe_m[0]`	$m_\phi = 0.30,\ g = 3.7 \times 10^{-6}$	$c\tau$ [m] (total width)
`ctau_probe_m[1]`	$m_\phi = 0.30,\ g = 1.11 \times 10^{-5}$	$c\tau$ [m] (total width)